

Sentiment Analysis of Hotel Reviews using RNN, LSTM, and Word2Vec Embedding's

6CS012 Final Portfolio Project – NLP Task

Student Name: Lam Pasang Lama

Student ID: 2408830

College: Herald College Kathmandu

Module: Artificial Intelligence and Machine Learning (6CS012)

Dataset: Hotel Reviews Dataset

Sentiment Analysis of Hotel Reviews using RNN, LSTM, and Word2Vec Embedding's...1

Contents

6CS012 Final Portfolio Project – NLP Task.....	1
1. Introduction	3
2. Data Preprocessing & Preparation.....	3
3. Model Architectures	3
4. Model Training and Evaluation.....	4
5. Visual Evidence (Screenshots)	4
6. Error Analysis.....	6
7. Conclusion and Future Work	6

1. Introduction

This project implements a sentiment analysis system using deep learning to classify hotel reviews. The goal is to compare the performance of a baseline Simple RNN, a standard LSTM, and a transfer learning approach using pre-trained Glove embedding's.

2. Data Preprocessing & Preparation

The `Hotel_Reviews.csv` dataset was processed to prepare it for deep learning:

- Labelling: Ratings were converted into binary sentiment classes where ratings ≥ 4 are Positive (0) and < 4 are Negative (1).
- Text Cleaning: Lowercasing, punctuation removal, and numeric filtering were applied.
- Lemmatization: Words were reduced to their root forms (e.g., "staying" to "stay") to improve model generalization.
- Tokenization & Padding: A vocabulary of 10,000 words was used, and sequences were padded to a length of 100.

```
# Labeling: Rating >= 4 is Positive (Label 0), < 4 is Negative (Label 1)
df['Label'] = df['Rating'].apply(lambda x: 0 if x >= 4 else 1)

print("Class Distribution:")
print(df['Label'].value_counts())
df.head()
```

```
... Class Distribution:
Label
0    15093
1     5398
Name: count, dtype: int64
```

	Review	Rating	Label
0	nice hotel expensive parking got good deal sta...	4	0
1	ok nothing special charge diamond member hilto...	2	1
2	nice rooms not 4* experience hotel monaco seat...	3	1
3	unique, great stay, wonderful time hotel monac...	5	0
4	great stay great stay, went seahawk game aweso...	5	0

3. Model Architectures

Three distinct architectures were implemented using the Keras Sequential API:

- Model 1: Simple RNN: A baseline model using a trainable embedding layer and a SimpleRNN layer.
- Model 2: Standard LSTM: An advanced architecture designed to capture long-term dependencies in the text.
- Model 3: Glove LSTM: A transfer learning model utilizing 50-dimensional pre-trained Glove embedding's to leverage semantic relationships from a larger corpus.

4. Model Training and Evaluation

The models were trained using the Adam optimizer and Binary Cross entropy loss.

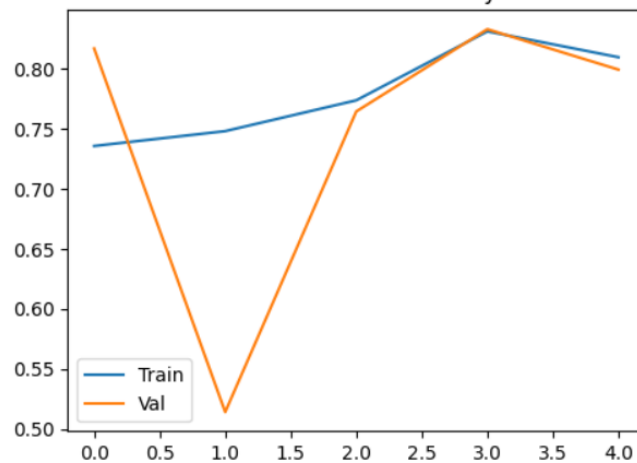
4.1. Performance Comparison Table

Model Type	Training Accuracy	Validation Accuracy	F1-Score (Negative)
Simple RNN	81.33%	77.13%	0.40
Standard LSTM	83.45%	79.15%	0.63
Word2Vec (Glove)	73.42%	75.00%	0.09

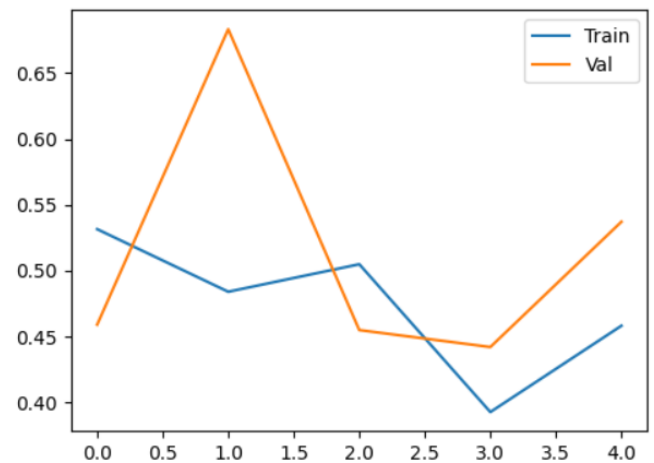
5. Visual Evidence (Screenshots)

To validate the model performance, the training history and classification metrics for the best model (Standard LSTM) are provided below:

Standard LSTM Accuracy



Standard LSTM Loss

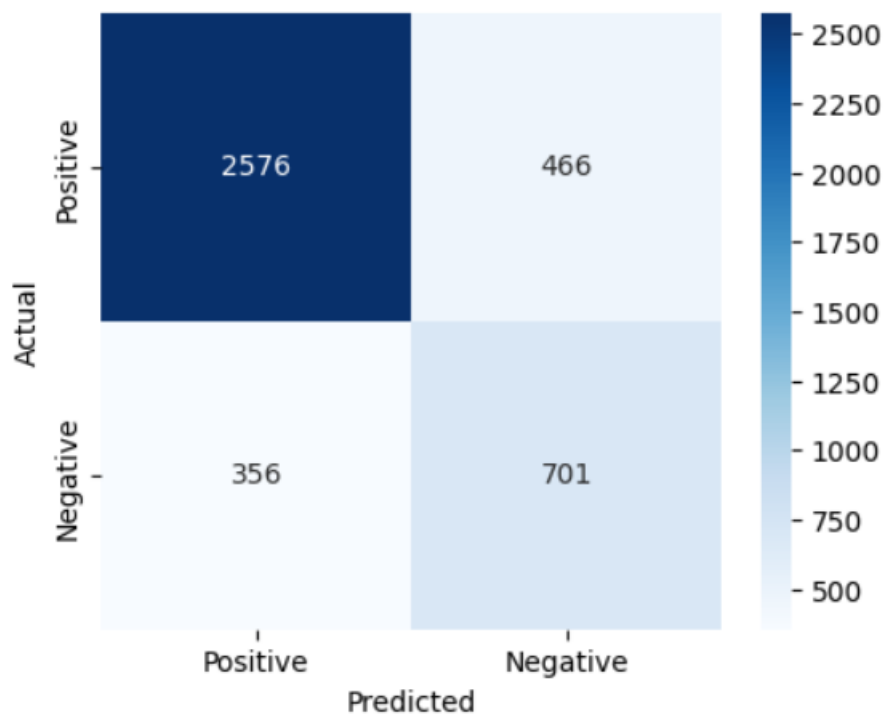


129/129 2s 17ms/step

--- Standard LSTM Classification Report ---

	precision	recall	f1-score	support
Positive	0.88	0.85	0.86	3042
Negative	0.60	0.66	0.63	1057
accuracy			0.80	4099
macro avg	0.74	0.76	0.75	4099
weighted avg	0.81	0.80	0.80	4099

Confusion Matrix: Standard LSTM



6. Error Analysis

By reviewing the Confusion Matrix, it is evident that while the model is highly accurate at identifying positive reviews, it occasionally misclassifies negative reviews. This is often due to:

- Mixed Sentiment: Reviews containing both praise and complaints (e.g., "Great room but bad service").
- Sarcasm: Phrases that are linguistically positive but contextually negative.

7. Conclusion and Future Work

The Standard LSTM (Model 2) provided the most effective results. The gating mechanism (Input, Forget, and Output gates) allows the LSTM to retain meaningful context over long sequences, solving the vanishing gradient problem found in Simple RNNs.

Future Work: To further improve accuracy, I would recommend:

- Implementing Bidirectional LSTMs to read context from both directions.
- Utilizing Attention Mechanisms to focus on key descriptive adjectives.
- Applying BERT (Transformer) models for state-of-the-art language understanding.